



A prediction method with uncertainty-aware for the remaining useful life and state of health of the proton exchange membrane fuel stack

Getnet Awoke Kebede^{a,b}, Fu-Kwun Wang^{a,*}, Bereket Haile Woldegiorgis^b

^a Department of Industrial Management, National Taiwan University of Science and Technology, Taipei City, 106335, Taiwan

^b Faculty of Mechanical and Industrial Engineering, Bahir Dar Institute of Technology, Bahir Dar University, Bahir Dar, Ethiopia

HIGHLIGHTS

- A new uncertainty-aware prognostic framework based on QsLSTM is proposed.
- Transfer learning is employed by using degradation-informed initialization.
- The proposed method offers RMSE as low as 0.0014 and PI's coverage >94 %.

ARTICLE INFO

Keywords:

Proton exchange membrane fuel
Remaining useful life prediction
Transfer learning
Uncertainty

ABSTRACT

Proton exchange membrane fuel cells (PEMFCs) are increasingly used in sustainable energy systems, but predicting their degradation under dynamic operating conditions remains difficult. This study introduces a new uncertainty-aware prognostic framework that combines a quantum-enhanced scalar long short-term memory (QsLSTM) model with transfer learning and joint supervision to generate prediction intervals. The model captures complex temporal degradation patterns from hybrid sensor and voltage data using shared linear embedding and variational quantum circuits embedded within scalar LSTM gates. A custom loss function improves point prediction accuracy, coverage probability, and interval sharpness simultaneously. To enhance model generalization in data-scarce situations, transfer learning is applied from a source to a target using degradation-informed initialization. Experimental results demonstrate that the proposed method achieves superior prediction accuracy (RMSE as low as 0.0014) and well-calibrated uncertainty bounds (coverage greater than 94 %), outperforming models such as BiLSTM, QLSTM, the linear layer-enhanced QLSTM, and Transformer. Additionally, real-time rolling RUL forecasts exhibit near-zero relative errors and narrow prediction intervals. The approach provides a scalable and robust solution for PEMFC health monitoring and lifespan extension.

1. Introduction

Fuel cell technology has gained widespread recognition for its ability to utilize hydrogen as a clean energy source [1]. In global efforts toward energy conservation and environmental sustainability, fuel cells are regarded as one of the most promising power generation technologies [2], drawing significant interest from academia and industry. Among the various types of fuel cells, PEMFCs have become a leading choice, offering high energy efficiency, excellent power density, quick start-up at room temperature, low noise, portability, and minimal greenhouse gas emissions [3]. Compared to other fuel cell technologies, such as alkaline, phosphoric acid, and molten carbonate fuel cells, PEMFCs demonstrate superior efficiency and adaptability [4]. Their usefulness

makes them suitable for commercial, industrial, and residential applications, especially in electric mobility, power generation, maritime transport, defense, and portable electronics [5]. Particularly, the transportation sector has embraced PEMFCs for powering eco-friendly vehicles, significantly contributing to the rapid growth of the electric vehicle (EV) market in the past decade [4–6].

Aligned with the Paris Agreement climate goals, the International Energy Agency (IEA) projects that EV sales will reach 250 million units, accounting for 35 % of global car sales, by 2030 [7]. The global EV market is expected to grow from USD 396.49 billion in 2024 to USD 620.33 billion by 2030. Despite these encouraging trends, PEMFCs face significant challenges related to performance degradation, which shortens their lifespan and increases maintenance costs [5,6,8].

* Corresponding author.

E-mail address: fukwun@mail.ntust.edu.tw (F.-K. Wang).

<https://doi.org/10.1016/j.jpowsour.2025.238844>

Received 19 August 2025; Received in revised form 2 November 2025; Accepted 12 November 2025

Available online 15 November 2025

0378-7753/© 2025 Elsevier B.V. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

Operational stresses such as load cycling and environmental conditions contribute to efficiency loss and premature failure. For example, while automotive PEMFCs are designed to last over 5000 h (up to 8000 h for cars and 25,000 h for buses), actual lifetimes often range between 2500 and 4130 h due to harsh conditions [9]. This durability gap remains a key barrier to the large-scale commercialization and industrial adoption of PEMFC systems [10]. To address these issues, prognostics and health management (PHM) systems have been developed as essential tools for extending the lifespan of PEMFCs and minimizing life cycle costs [11]. PHM includes real-time monitoring, state of health (SOH) estimation, fault diagnosis, and remaining useful life (RUL) prediction [12]. Accurate RUL forecasting supports preventive maintenance, enhances system reliability, and improves safety. Additionally, understanding degradation patterns is crucial for optimizing system management and enhancing performance [13]. Three main categories of RUL prediction methods have been established for PEMFCs: model-based, data-driven, and fusion approaches [14,15]. Model-based techniques depend on physical, empirical, or semi-empirical formulations grounded in the system's geometry, materials, and aging mechanisms. These methods provide physical interpretability and require minimal data but are constrained by the complexity and specificity of fuel cell degradation models [16]. Conversely, data-driven models utilize machine learning and statistical techniques to directly infer degradation trends from operational data, eliminating the need for detailed system knowledge [17,18]. Fusion models combine both model-based and data-driven strengths to enhance RUL prediction and condition monitoring [19,20]. While these hybrid approaches improve predictive accuracy, they are computationally intensive and sensitive to parameter settings, which may limit their scalability [21].

Recently, deep learning has emerged as a dominant paradigm for fuel cell prognostics due to its flexibility and powerful pattern recognition capabilities [22]. Supervised learning models, trained on features such as voltage, current, temperature, and charge, have demonstrated strong performance in estimating state of health (SOH) and RUL [23]. Variants of recurrent neural networks (RNNs), particularly long short-term memory (LSTM), gated recurrent unit (GRU), and convolutional neural network (CNN), have proven effective in capturing temporal dependencies in PEMFC degradation data [24–26]. For instance, both single- and multi-step LSTM models have been used to forecast short-term voltage loss [26], and LSTM architectures have shown superior accuracy under dynamic operating conditions [27]. GRU-based models have also demonstrated robustness in predicting the RUL of hydrogen fuel cells [28], while CNNs combined with the random forest (RF) algorithm have improved SOH estimation under dynamic load conditions [29]. Hybrid frameworks that integrate CNNs and RNNs have further enhanced prediction performance. For example, a 2D CNN-GRU hybrid model [30], a CNN-LSTM pipeline [31], and a BiLSTM-GRU-ESN ensemble [32] have all been proposed to improve feature extraction and reduce training data requirements. In particular, BiLSTM models combined with self-attention mechanisms [18] or transfer learning [33] have effectively captured bidirectional temporal patterns and reduced data dependency. Furthermore, the quantum-enhanced LSTM (QLSTM) architecture has been explored for improving the RUL prediction accuracy [34].

Beyond RNN-based methods, emerging deep learning models based on Transformer architectures and state-space representations have shown promising results. For example, the temporal fusion transformer (TFT) has demonstrated superior performance in capturing complex degradation trends [35], while Informer models have enabled accurate multi-step RUL prediction using generative inference [36]. Health indicators such as stack voltage, power output, current density, and internal resistance tend to be static and often become inaccurate under dynamic conditions where operating environments change frequently [37–40].

Most existing studies focus solely on point-based SOH or RUL predictions, providing limited insight into the uncertainty underlying these

estimates. The absence of probabilistic information constrains robust decision-making in practical operational contexts. Only a few recent studies—Bayesian GRU [41], Bayesian LSTM [42], Bayesian CNN with attention [43], and transformer models combined with Gaussian process regression [44]—have emphasized the importance of uncertainty-aware prognostics in bridging this gap. However, these methods primarily concentrate on estimating uncertainty and often fail to deliver a comprehensive assessment of prediction intervals.

To address this limitation, the present study adopts a prediction interval-based framework for uncertainty quantification. The proposed approach simultaneously learns the lower, point, and upper bounds of degradation forecasts, enabling explicit estimation of predictive uncertainty while enforcing interval consistency through soft constraints. A tailored loss function is designed to balance point-wise accuracy, interval coverage, and sharpness. To assess the reliability of the uncertainty estimates, evaluation metrics such as coverage probability and normalized average interval width (NAIW) are employed, ensuring both interpretability and robustness of the prognostic outcomes.

Real-world applications of fuel cells, such as in electric vehicles, are subjected to highly dynamic operating environments that static health indicators cannot adequately capture [45]. Therefore, it is necessary to develop methods capable of assessing a fuel cell's health by providing dynamic health metrics and predicting its degradation—an essential requirement for the large-scale commercialization of fuel cell technology. In this context, only a limited number of studies have employed the relative voltage drop rate (RVDR) as a health indicator for evaluating the condition of PEMFCs.

Here, RVDR is defined as: $\Delta U = (U_0 - U)/U_0$, where U_0 is the initial voltage (healthy condition) and U is the current voltage of the PEMFC stack. Technically, the RVDR normalizes performance loss across stacks, enabling better generalization under different initial conditions. Also, as a dimensionless and scale-independent metric, RVDR provides a value that is less sensitive to operational fluctuations, allowing it to better reflect the actual aging of the fuel cell under dynamic loads, which can improve the model's sensitivity to early degradation. In other words, the RVDR captures degradation of a fuel cell under changing real-world operating conditions by comparing current voltage to an initial, ideal voltage (healthy condition) for the same operating condition, providing a more comprehensive measure of a PEMFC's state of health. This dynamic approach is crucial for accurate prognostics, especially in applications like public transport, where voltage can fluctuate frequently. A higher RVDR indicates greater voltage loss, and this indicates a lower state of health.

In addition to the above-mentioned limitations, the possibility of a real-time PEMFC's RUL prediction based on the current operating conditions (real-values) is still a gap in the domain of PEMFC's life analysis. However, in real-world applications, long-term forecasts based on historical observations are often required, and continuous access to real future values is not possible [46]. Thus, in this paper, we employed a prediction method that enables a real-time rolling prediction of RUL by continuously updating inputs and identifying the time step when predicted voltage falls below a predefined end-of-life (EOL) threshold, providing interpretable prognostics with quantified confidence.

In parallel, recent breakthroughs in quantum computing offer new opportunities for advancing PEMFC prognostics. Quantum-enhanced neural networks, such as QLSTM [47,48], integrate quantum circuits with classical architectures to improve learning capacity and uncertainty modeling. By leveraging quantum entanglement and parallelism, these models can efficiently process high-dimensional, nonlinear data and capture complex variable interactions [49].

Motivated by these opportunities and challenges, this paper proposes a PEMFC SOH and RUL prediction framework. Our approach integrates scalar-LSTM architectures with quantum algorithms, a linear embedding layer, and transfer learning techniques. This hybrid design addresses key limitations in computational efficiency, generalization, and feature representation, enabling a robust, scalable, and real-time PEMFC

prognostic system suitable for deployment in real-world settings. The main contributions of this paper are as follows:

1. A quantum-enhanced scalar LSTM model is developed to effectively extract temporal patterns from hybrid voltage and sensor data.
2. A new prediction-interval approach jointly estimates the lower, point, and upper bounds of degradation, using a custom loss function that balances accuracy, coverage, and sharpness. Uncertainty quality is evaluated using coverage probability and normalized average interval width.
3. The prediction method enables real-time rolling predictions of RUL by continuously updating inputs and identifying the time step when predicted voltage falls below a predefined EOL threshold, providing interpretable prognostics with quantified confidence.

The rest of the paper is organized as follows: Section 2 describes the dataset and research methods. Section 3 presents the analysis results and discussion. Finally, Section 4 concludes the paper.

2. Material and methods

2.1. Dataset

To evaluate the effectiveness of the proposed model, this study utilizes the publicly available datasets, FC1 and FC2, which were released as part of the IEEE 2014 data challenges. These datasets were generated through controlled durability testing of proton exchange membrane fuel cells using a laboratory test bench named FR CNRS 3539, operated by the FCLAB research federation in France. The summary of the characteristics of the dataset is given in Table 1.

The variables recorded during the experiment are critical in understanding and modeling the voltage degradation behavior of PEMFC stacks over time. The degradation trends are illustrated in Fig. 1, where the voltage decline across operating time can be observed for both datasets. This pattern serves as the primary signal for predicting the SOH and of the stacks. For more detailed information about the experimental setup and data acquisition process, readers are referred to Ref. [50].

2.2. Data preprocessing

By applying the random forest algorithm [51], eleven variables out of all 24 recorded variables were identified as they have a significant impact on the target variable and are prioritized as the most important input features for model predictions (see Fig. 1). For instance, the inlet and outlet temperatures of water, air, and H₂ gas are very important. The performance of the PEMFC is highly sensitive to temperature, which means that temperature control is an indispensable part of the normal operation of PEMFC [52]. Hence, temperature is considered as one of the critical parameters affecting the performance of PEMFC, such as electrochemical activity, maximum theoretical voltage and mass diffusion coefficient.

Table 1
Overview of the FC1 and FC2 datasets.

Category	Parameters	FC1	FC2
General info	Number of cells	5	5
	Active area per stack	100 cm ²	100 cm ²
Operating conditions	Operating temperature	55 °C	55 °C
	Current load profile	Constant (70 A)	Dynamic (63–77 A)
	Temperature range	20 °C–80 °C	20 °C–80 °C
	Relative humidity (RH)	0 %–100 %	0 %–100 %
	Hydrogen flow rate	0–30 L/min	0–30 L/min
	Air flow rate	0–100 L/min	0–100 L/min
	Gas pressure	0–2 bars	0–2 bars
Data characteristics	# of recorded variables	~24 variables including electrical, thermal, fluidic, and environmental	Same as FC1
	Total Data Points	143,861	127,369
	Degradation Pattern	Gradual voltage decay over time	More variable degradation trend due to dynamic load

Selecting an appropriate starting point for model training is critical in battery and fuel cell prognostics. It helps eliminate irrelevant data from early-life stages, reduces training time, and improves computational efficiency. Several studies [33,35] emphasize the importance of identifying the knee point, the moment when fuel cell degradation transitions from a slow to a rapid decline. Recognizing this inflection point offers valuable guidance for determining the optimal starting point for model training. To address this, the current study employs the Double-Bacon method to detect the knee points in both FC1 and FC2 datasets (see Fig. 2). This approach enables more precise selection of training windows, improving model performance in both the source domain and during fine-tuning in the target domain.

The double-Bacon method (DBM) models the state of health as a piecewise function with smooth transitions. It is defined as:

$$\text{SOH} = \alpha_0 + \alpha_1(x - x_0) + \alpha_2(x - x_0) \tanh[(x - x_0)/\gamma] + \alpha_3(x - x_1) \tanh[(x - x_1)/\gamma] + Z \quad (1)$$

where x is the time in hour, Z is the residual error assumed to follow a normal distribution with mean 0 and constant variance, α_0 represents the intercept for the leftmost segment, α_1 , α_2 , and α_3 control the slopes of the piecewise segments, and γ regulates the sharpness of the transition between phases. This formulation ensures a smooth yet distinct capture of the degradation dynamics, making it well-suited for identifying the onset of accelerated degradation and guiding the model's learning focus.

2.3. The prognostic model for PEMFC stack based on the QsLSTM model

A prognostic framework for estimating the SOH and RUL of PEMFC stacks using a quantum-enhanced scalar long short-term memory (QsLSTM) model is presented. The architecture combines classical deep learning with emerging quantum computing techniques to extract degradation patterns from hybrid input data, including stack voltage and sensor measurements. In Fig. 3, the pipeline includes input sequence generation, data preprocessing, model initialization, transfer learning, and dynamic SOH and RUL estimation. The details of each step are given as the following subsections (see Fig. 4).

The QsLSTM framework is constructed by integrating scalar LSTM (sLSTM) and variational quantum circuits (VQCs). The sLSTM model is a recent extension of the LSTM architecture designed to improve memory control, gradient stability, and computational scalability [53]. It introduces exponential gates, normalization states, and stabilizer units to better capture long-term dependencies and provide numerical robustness, particularly valuable in small and noisy time-series datasets typical of PEMFC systems. The scalar LSTM structure, adapted from the extended-LSTM (xLSTM) architecture, includes exponential activation for input and forget gates, normalization for cell state updates, and residual stacking of layers for enhanced representational power. It is particularly effective for small-sample, time-series prediction tasks and has been demonstrated to perform competitively with state-of-the-art architectures such as transformers and state-space models.

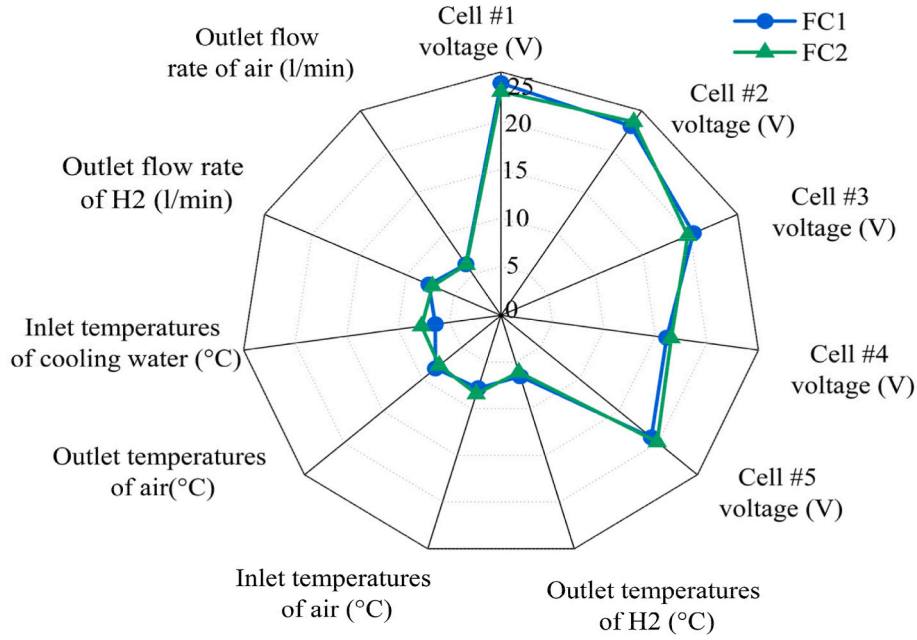


Fig. 1. Feature importance ranking and selection.

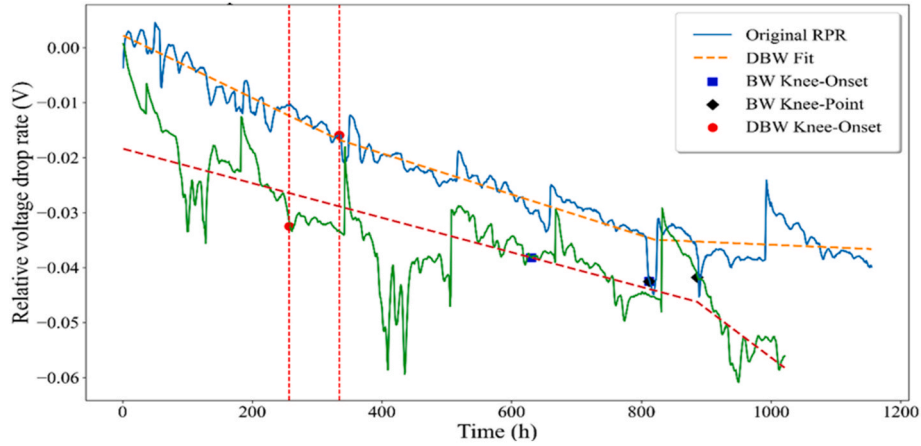


Fig. 2. Knee point and onset detection using the DBW model across FC1 and FC2 datasets.

The operational components of the sLSTM are defined in Eqs. (2)–(16). The cell state c_t serves as the long-term memory of the network, updated by the input gate i_t , which controls the contribution of the current input and previous hidden state, and the forget gate (f_t), which determines how much of the previous memory (c_{t-1}) is retained. The output gate o_t regulates the information passed from the cell state to the hidden state (h_t). To ensure numerical stability when using exponential activations in the input and forget gates, a normalization mechanism is introduced, as shown in Eq. (5). Additionally, Eqs. (15) and (16) define the stabilized gates $\tilde{i}_t$ and $\tilde{f}_t$, which are rescaled versions of i_t and f_t to prevent numerical overflow.

The overall sLSTM update combines exponential gating, normalized memory updates, and traditional LSTM operations. Specifically, the internal candidate state z_t is calculated via an affine transformation of the input x_t and the previous hidden state h_{t-1} , passed through a non-linear activation φ . The normalized memory accumulator n_t ensures that updates to h_t are stable and interpretable.

$$c_t = f_t c_{t-1} + i_t z_t \quad (2)$$

$$\tilde{z}_t = W_z^T x_t + r_z h_{t-1} + b_z \quad (3)$$

$$z_t = \varphi(\tilde{z}_t) \quad (4)$$

$$n_t = f_t n_{t-1} + i_t \quad (5)$$

$$h_t = o_t \tilde{h}_t \quad (6)$$

$$\tilde{h}_t = c_t / n_t \quad (7)$$

$$i_t = \exp(\tilde{i}_t) \quad (8)$$

$$\tilde{i}_t = W_i^T x_t + r_i h_{t-1} + b_i \quad (9)$$

$$f_t = \sigma(\tilde{f}_t) \quad (10)$$

$$\tilde{f}_t = W_f^T x_t + r_f h_{t-1} + b_f \quad (11)$$

$$o_t = \sigma(\tilde{o}_t) \quad (12)$$

$$\tilde{o}_t = W_o^T x_t + r_o h_{t-1} + b_o \quad (13)$$

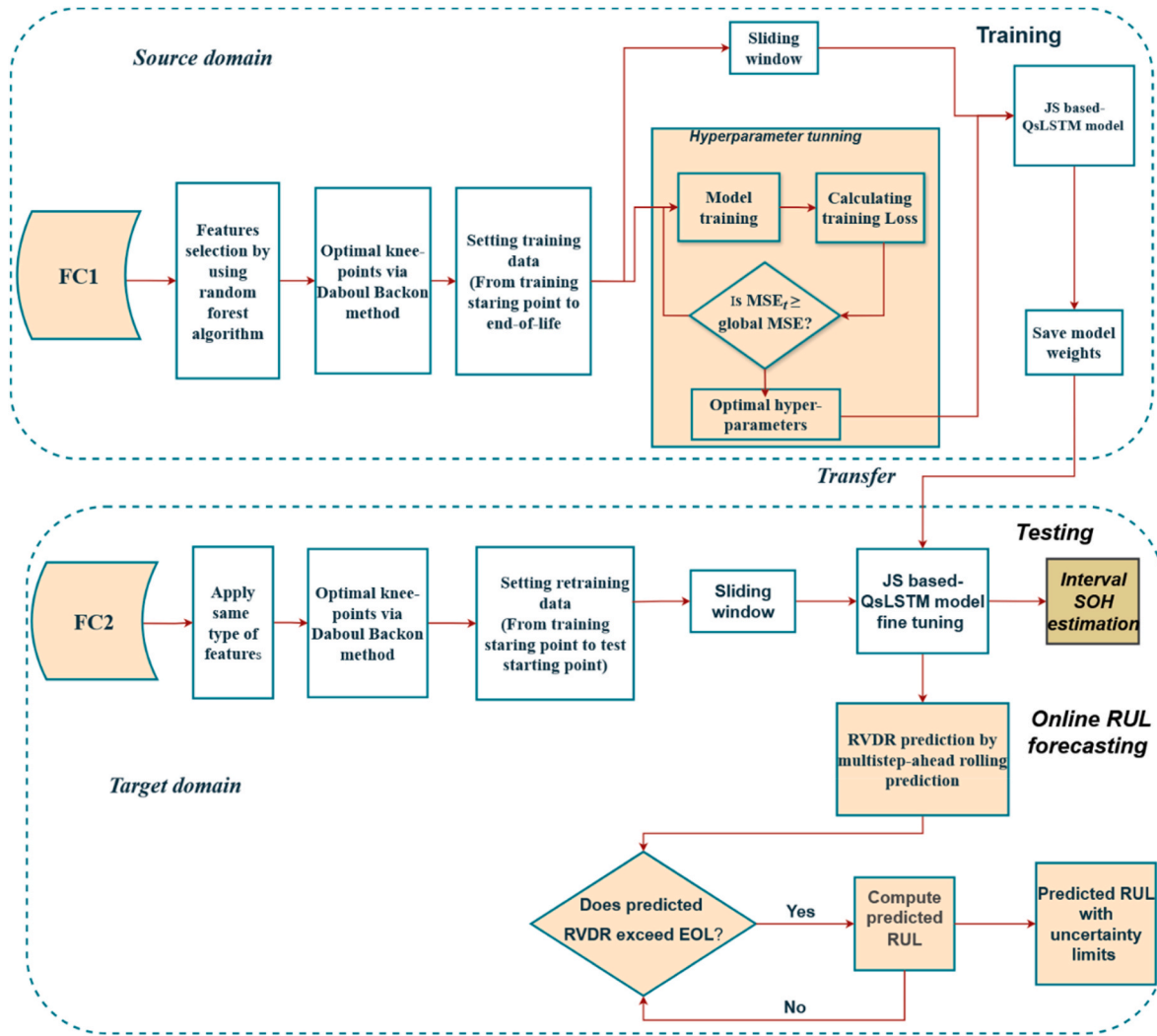


Fig. 3. The proposed PEMFC SOH and RUL prediction model.

$$m_t = \max(\log(f_t) + m_{t-1} - m_t) \quad (14)$$

$$\tilde{f}_t = \exp(\log(\tilde{f}_t) - m_t) = \exp(\tilde{f}_t - m_t) \quad (15)$$

$$\tilde{f}_t = \exp(\log(f_t) + m_{t-1} - m_t) \quad (16)$$

where x_t denotes the current input, h_{t-1} the previous hidden state, and c_{t-1} the last state cell. Variables W^* , r^* , b^* represent the weight matrices, recurrent connections, and biases for each gate, respectively. The term $\tilde{f}_t$ and $\tilde{f}_t$ are the stabilized forms of the input and forget gates, while m_t helps regulate the exponential operations. This architecture enhances stability and memory dynamics, making sLSTM particularly suitable for complex time-series modeling tasks such as PEMFC degradation prediction.

Quantum computing, based on principles such as superposition, entanglement, and unitary evolution, offers a fundamentally different paradigm for information processing [47,49]. Unlike classical bits, quantum bits (qubits) can exist in multiple states simultaneously, allowing quantum algorithms to represent and manipulate information in high-dimensional spaces with exponential capacity. Although current noisy intermediate-scale quantum (NISQ) devices are still limited by hardware constraints, such as short coherence times and limited qubit counts, they have shown promise in hybrid quantum-classical applications, especially through variational quantum algorithms [47,55,56].

In the QLSTM architecture (see Fig. 6), Variational quantum circuits

(VQCs) act as powerful nonlinear optimizers within a hybrid framework. A VQC consists of parameterized quantum gates, mainly rotations and entanglements, whose parameters are optimized using classical algorithms. Classical input data, such as battery sensor readings (e.g., current, voltage, temperature, and internal resistance), along with historical features, are first encoded into quantum states using methods like amplitude or angle encoding. The VQC then applies a unitary transformation in the quantum Hilbert space, capturing both individual feature impacts and their complex interdependencies. This makes the VQC a nonlinear function approximator by nature. Each VQC processes its input through three stages: encoding, quantum transformation, and measurement. The final output, a classical scalar or vector, is obtained from quantum expectation values and represents a learned nonlinear mapping of the input. This output can be used directly for prediction or integrated into larger models like LSTMs.

The role of the VQC is to minimize a loss function, such as the mean squared error between predicted and actual battery SOH. Its parameters are iteratively fine-tuned using classical optimizers (e.g., gradient-based or gradient-free methods). VQCs can model cross-feature relationships through entanglement. For example, they can capture how temperature modulates the effect of current on battery degradation, something that classical models often struggle with unless they are deeply layered or feature-engineered. VQCs handle such complex, condition-dependent interactions more efficiently.

The model computations of a QsLSTM cell proceed as follows:

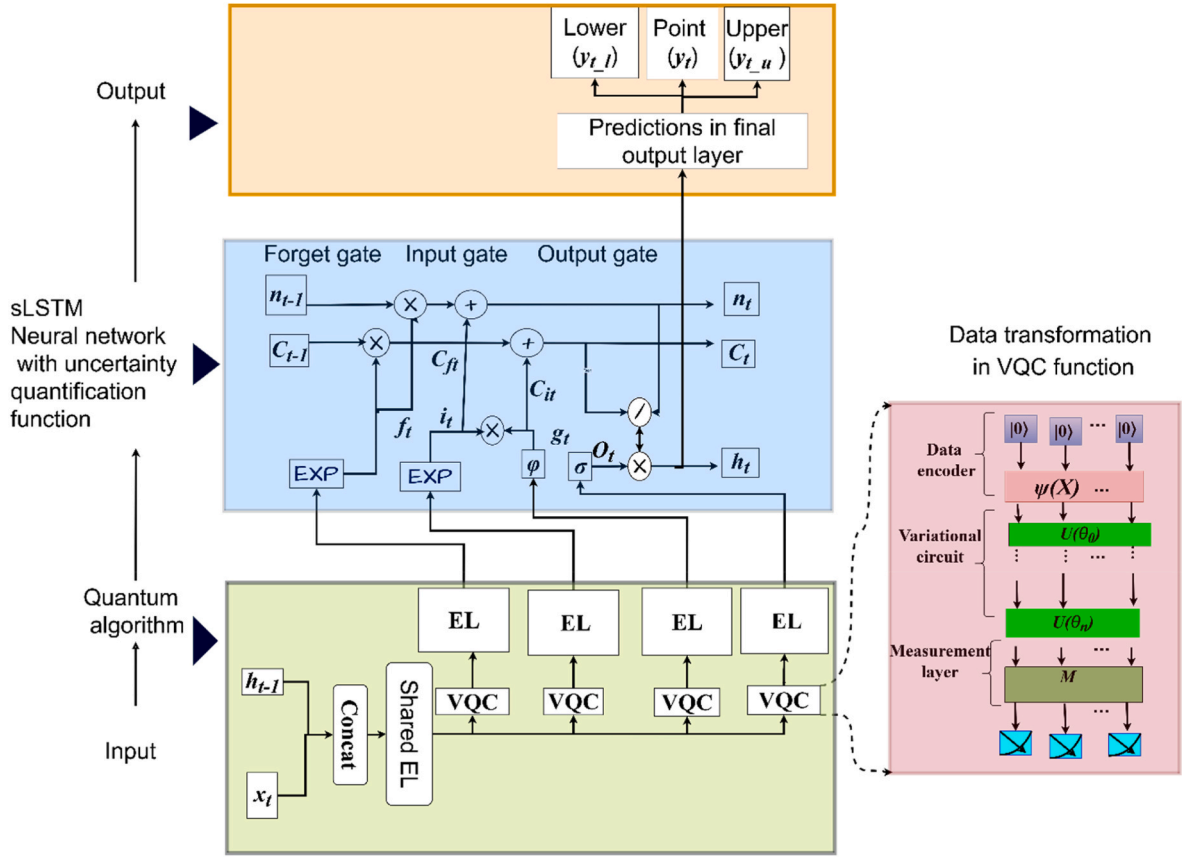


Fig. 4. The structure of a shared linear embedding layer-based QsLSTM cell (Left) with a VQC function (Right).

$$z_t = \text{Shared linear embedding layer}(\text{concat}(X_t, h_{t-1})) \quad (17)$$

$$f_t = \exp(W(VQC_1(z_t) + b_f)) \quad (18)$$

$$i_t = \exp(W(VQC_2(z_t) + b_i)) \quad (19)$$

$$g_t = \tan h(W(VQC_3(z_t) + b_g)) \quad (20)$$

$$C_t = f_t * C_{t-1} + i_t * g_t \quad (21)$$

$$O_t = \sigma(W(VQC_4(z_t) + b_o)) \quad (22)$$

$$h_t = (\tan h(C_t) * \tan h(O_t)) \quad (23)$$

$$y_t = (\tan h(C_t) * (O_t)) \quad (24)$$

The shared linear embedding operation is formally defined as:

$$y = Wx + b \quad (25)$$

where W denotes the weight matrix, x is the input tensor, and b is the bias term. This operation maps the high-dimensional hybrid input into a lower-dimensional latent space suitable for quantum circuits, while preserving meaningful relationships across features.

To support real-time operation, the QsLSTM model performs online rolling predictions. At each time step, it updates its input with the most recent measurements and dynamically estimates the RUL by identifying the time when the predicted voltage falls below a predefined end-of-life (EOL) threshold. This enables timely and informed prognostic decision-making. Overall, the proposed QsLSTM framework offers a hybrid solution that combines the memory efficiency and modeling capabilities of sLSTM with the expressiveness of quantum computing. The use of shared embeddings and parameter-efficient design ensures compactness, while

compatibility with NISQ hardware makes the approach both forward-looking and practically viable. This architecture enables accurate, interpretable, and uncertainty-aware degradation prediction for PEMFC applications.

2.4. Transfer learning via model weight initialization and fine-tuning

Transfer learning is a powerful technique that uses knowledge and features learned from a source domain to improve the training and performance of a model in a target domain. By leveraging existing knowledge, this method increases predictive accuracy and enhances the model's ability to generalize across different environments [57,58]. The main goal of transfer learning is to utilize large amounts of data from the source domain to train a strong model while dealing with data scarcity issues in the target domain. This allows the model to perform similar tasks in slightly different scenarios.

Mathematically, transfer learning can be formulated as an optimization problem where the goal is to minimize the loss function of the target task while incorporating knowledge from a pre-trained model. In general, the optimization procedure consists of three stages: (i) source domain model pretraining, (ii) knowledge adaptation through weight transfer, and (iii) fine-tuning and retraining on the target domain.

Let D_s , x_s , y_s , $(f(x_s; \theta))$, L , and θ_s represent the dataset, input features, labels, model function, loss function, and learned parameters in the source domain. Similarly, D_t , x_t , y_t , $(f(x_t; \theta))$, θ_t , R and θ^* denote the corresponding components in the target domain where R is a regularization term that penalizes deviations from the pre-trained weights. The optimization objectives are given by:

$$\theta_s = \arg \min_{(x_s, y_s) \in D_s} \sum L(f(x_s; \theta), y_s) \quad (26)$$

$$\theta_t = \text{arg min}_{(x_t, y_t) \in D_t} L(f(x_t; \theta), y_t) + \lambda R(\theta, \theta_s) \quad (27)$$

$$\theta^* = \text{arg min}_{(x_t, y_t) \in D_t} L(f(x_t; \theta), y_t) \quad (28)$$

To implement this in practice, we designed a flexible transfer learning strategy for our structured QsLSTM model. The strategy supports partial or full weight transfer from a source model trained on the FC1 fuel cell stack dataset. Specifically, the transfer learning function enables loading pre-trained weights from a saved checkpoint. It provides an option to freeze all model parameters, effectively turning the model into a fixed feature extractor, or to allow fine-tuning of the transferred layers for domain adaptation. The function also logs missing or unexpected keys to ensure compatibility between model architectures.

In our experiments, we applied this transfer learning framework to fine-tune the model on the FC2 dataset, where labeled samples are limited. The source model, trained on FC1, provides a strong initialization point by encoding prior degradation patterns. Fine-tuning then allows adaptation to FC2's specific degradation behavior, thereby improving predictive accuracy and robustness. This approach balances knowledge reuse and model flexibility, achieving better generalization in the target domain.

2.5. Hyperparameter optimization

Bayesian optimization stands out as a powerful tool for addressing challenges in hyperparameter tuning within machine learning [59]. In this study, it was employed to systematically refine four key hyperparameters of the QLSTM model, which incorporates a linear embedding layer: epochs, batch size, lookback, and dropout rate. The optimization process was guided by the mean squared error (MSE) function, chosen as the evaluation metric, to identify an optimal hyperparameter configuration that enhances predictive performance. The MSE test function operates within the interval [0, 1], with an MSE value of 0.1 serving as the stopping criterion for the iterative optimization process.

2.6. Prediction intervals

To improve both the interpretability and reliability of SOH and RUL predictions for safety-critical applications, such as energy storage systems (e.g., fuel cells, lithium-ion batteries), it is essential to incorporate uncertainty quantification (UQ) into the predictive modeling process [60]. Rather than producing only a single point estimate, the proposed model generates three outputs for each input sequence: a lower bound, a central point prediction, and an upper bound. These outputs form a prediction interval that captures the range of reasonable future values, providing a probabilistic view of model confidence. Conventional uncertainty estimation methods, such as Monte Carlo (MC) dropout and Bayesian ensemble, often derive prediction intervals from the distribution of point estimates. However, these approaches may lead to inconsistent interval ordering or under-coverage, as they do not explicitly supervise the interval endpoints during training. This limitation reduces the interpretability and robustness of the resulting uncertainty bounds [61].

To address this challenge, the model presented in this study adopts a joint supervision-based prediction interval learning strategy [62]. This approach directly and simultaneously trains the lower, central, and upper outputs by minimizing a custom interval loss function that enforces three key objectives: (i) point prediction accuracy, (ii) high coverage probability (i.e., the proportion of true values falling within the interval), and (iii) narrow and sharp intervals. The loss incorporates soft constraints to maintain a proper interval during the training process.

The prediction interval is produced by three parallel fully connected heads at the output layer of the QsLSTM model. To ensure valid interval

formation and avoid degenerate or overlapping bounds, the model leverages the softplus function and a small positive margin to enforce non-crossing behavior in the interval outputs. This design guarantees that the resulting intervals are not only mathematically consistent but also interpretable and practically useful.

To assess the quality of the uncertainty estimates, two primary metrics are used: coverage percentage, which quantifies how often the true target values lie within the predicted intervals, and normalized average interval width, which reflects the trade-off between interval sharpness and coverage. These metrics provide a comprehensive evaluation of the model's uncertainty estimation capability in real-world degradation forecasting and RUL applications.

To effectively supervise the model's outputs, we design a custom joint loss function termed interval loss. This loss function integrates four key components: (i) point prediction accuracy, measured by mean squared error; (ii) a coverage penalty, which penalizes instances where the ground truth falls outside the predicted interval; (iii) a sharpness penalty, which encourages narrower, more informative prediction intervals; and (iv) a constraint violation penalty, which enforces logical consistency by penalizing invalid (e.g., reversed) interval bounds. This composite objective guides the model to produce prediction intervals that are both accurate and compact, while faithfully capturing the underlying uncertainty. The mathematical formulation of this uncertainty-aware loss is provided in Eqs. (5)–(11). Let y_t , $\hat{y}_t^{\text{point}}$, $\hat{y}_t^{\text{lower}}$, $\hat{y}_t^{\text{upper}}$ be the true target value, central estimate, and lower estimates, which is described as $\hat{y}_t = [\hat{y}_t^{\text{lower}}, \hat{y}_t^{\text{upper}}]$ and with the central estimate $\hat{y}_t^{\text{point}}$. The loss function the interval learning is defined as:

To ensure that the prediction interval is valid and ordered, we apply soft constraints during output transformation:

$$\hat{y}_t^{\text{point}} = \hat{y}_t^{\text{lower}} + \text{softplus}(p_t - \hat{y}_t^{\text{lower}}) + \varepsilon \quad (29)$$

$$\hat{y}_t^{\text{upper}} = \hat{y}_t^{\text{lower}} + \text{softplus}(u_t - \hat{y}_t^{\text{point}}) + \varepsilon \quad (30)$$

where p_t and u_t are intermediate outputs of the network, and $\varepsilon > 0$ is a small constant to avoid zero-width intervals.

The total loss (L_{interval}) is a joint supervision loss composed of four terms, namely point prediction loss (MSE), interval penalty, sharpness penalty, and constraint violation penalty.

$$\begin{aligned} L_{\text{interval}} = & (1 - \lambda) \left(\frac{1}{N} \sum_t (\hat{y}_t^{\text{point}} - y_t)^2 \right) + (\lambda) \left(\frac{1}{N} \sum_t [\max(0, y_t \right. \\ & \left. - \hat{y}_t^{\text{upper}}) + \max(0, \hat{y}_t^{\text{lower}} - y_t)] \right) + \gamma \left(\frac{1}{N} \sum_t (\hat{y}_t^{\text{upper}} - \hat{y}_t^{\text{lower}}) \right) \\ & + (\omega) \left(\frac{1}{N} \sum_t [\max(0, \hat{y}_t^{\text{lower}} - \hat{y}_t^{\text{point}}) + \max(0, \hat{y}_t^{\text{point}} - \hat{y}_t^{\text{upper}})] \right) \end{aligned} \quad (31)$$

where the $\lambda \in [0, 1]$, $\gamma > 0$ and $\omega > 0$ are parameters denoting the trade-off point accuracy and interval penalty, weight for sharpness, and penalty for constraint violation.

2.7. Performance metrics

To comprehensively illustrate the efficacy of the proposed model, three metrics are employed: mean absolute percent error (MAPE), and root mean square error (RMSE). Evaluation is performed on model pretraining using the FC1 stack dataset and model fine-tuning using a small dataset of the FC2 stack to monitor model behavior. Also, model performance evaluation is performed on the SOH estimation ability of the trained model using the test data of the FC2 stack for testing the prediction accuracy of the proposed SOH and RUL predictive model. The following metrics are computed at each epoch during model training and FC2 stack SOH estimation.

$$MAPE = \frac{\sum_{i=1}^n |RVDR_t - \widehat{RVDR}_t|}{n} * 100\% \quad (32)$$

$$RMSE = \sqrt{\frac{\sum_{i=1}^n (RVDR_t - \widehat{RVDR}_t)^2}{n}}, \quad (33)$$

Similarly, the coverage probability (CP) and normalized average interval width (NAIW) are used to provide a qualitative assessment of the reliability and efficiency of the uncertainty estimates.

$$CP = \left(\frac{1}{N} \sum_t (\hat{y}_t^{lower} \leq y_t \leq \hat{y}_t^{upper}) \right) \quad (34)$$

$$NAIW = \left(\frac{1}{N} \sum_t \frac{\hat{y}_t^{upper} - \hat{y}_t^{lower}}{\max(y) - \min(y)} \right) \quad (35)$$

where the $RVDR_t$ and $\widehat{RVDR}_t$ denote the observed and predicted relative voltage drop rate, respectively, N denotes the number of data points, and t stands for the timestep in hours. MAPE and RMSE are used in the testing process to evaluate the model's performance. Similarly, the absolute error (AE) and relative error (RE) are used to measure the performance of rolling online RUL prediction.

$$AE = |RUL_{actual} - \widehat{RUL}_{predicted}| \quad (36)$$

$$RE = \frac{|RUL_{actual} - \widehat{RUL}_{predicted}|}{RUL_{actual}} \times 100\% \quad (37)$$

$$RA = 1 - \left(\frac{|RUL_{actual} - \widehat{RUL}_{predicted}|}{RUL_{actual}} \times 100\% \right) \quad (38)$$

3. Results and discussion

In this section, the results of the experiments are presented and discussed in detail, including the performance of the proposed model during training on the FC1 dataset, fine-tuning on the FC2 dataset, model testing for SOH estimation, and the model's ability for online RUL predictions. For a fair performance comparison, previously introduced related models, such as bi-directional LSTM with attention mechanism (BiLSTM-AM) [56], quantum long short-term memory (QLSTM) [44], and linear embedding layer-based QLSTM (LEQLSTM) [34], and Transformer were used as baseline models.

3.1. Model configuration and training progress

The model configurations and training procedure are summarized in Table 2. All models are trained for 200 epochs using stochastic gradient descent, suitable for the relatively small FC2 dataset. The proposed QsLSTM model uses a lookback window of 2–5, a batch size of 65, and a learning rate between 0.001 and 0.01, with a fixed dropout rate of 0.005. It incorporates a quantum circuit design (CD) consisting of multiple configurations, including 9 qubits with 2 layers, 7 qubits with 3 layers, and 5 qubits with 4 layers, structured to capture different levels

of quantum representational depth. Additionally, the QsLSTM includes 5 scalar LSTM blocks (B) and 1 attention head (H), which enhance temporal modeling and feature focus. Compared to baseline models (BiLSTM-AM, QLSTM, and LEQLSTM), the QsLSTM uniquely integrates quantum processing and scalar-LSTM components to improve the prediction of SOH and RUL in PEMFC stacks.

All experiments were conducted using Python 3.12 with PyTorch and the quantum-inspired machine learning library PennyLane (v0.3), executed within the Spyder 3.7.3 environment. Computations were performed on a local workstation equipped with an Intel® Core™ i9-9900K CPU @ 3.60 GHz, 16 GB RAM, NVIDIA GeForce GTX 1060 GPU, and Windows NT x64 operating system with a 1 TB hard disk.

To achieve both accurate forecasting and reliable uncertainty quantification, the QsLSTM model is trained using a joint supervision loss function. This composite loss integrates four components: (i) MSE for point predictions, (ii) a penalty for coverage errors when true values fall outside the predicted intervals, (iii) a sharpness term to discourage overly wide intervals, and (iv) a constraint loss to enforce proper interval ordering (i.e., lower < point < upper). The overall loss is calculated as a weighted sum of these terms, with dynamic scheduling of the weighting parameters (λ , γ , α) throughout training to balance predictive accuracy and interval quality. This design enables the model to generate both precise forecasts and well-calibrated prediction intervals.

The training progress of the proposed model, including RMSE and total loss curves during pretraining on the FC1 dataset, is illustrated in Fig. 5. As shown, the model converges rapidly within the first 20 epochs, with the total loss steadily decreasing and stabilizing near zero, indicating effective optimization. The R^2 score quickly increases and remains above 0.95, demonstrating a strong correlation between predicted and actual values. Additionally, the MAPE (%) curve drops sharply in the early epochs and stabilizes at a low value, confirming consistent improvements in relative prediction accuracy. These results highlight the model's fast convergence, high precision, and strong generalization during training.

Furthermore, a comparative analysis of computational resource use and training efficiency is conducted to clarify the trade-offs between model performance and computational cost. Table 3 compares the computational time and predictive accuracy of the proposed QsLSTM model with other benchmark models, including BiLSTM-AM, Transformer, and the quantum-enhanced variants (QLSTM and LEQLSTM), under different configurations of qubits and layers. The results indicate that incorporating quantum circuits into the LSTM framework significantly enhances prediction accuracy and uncertainty quantification while only moderately increasing computational time. The QsLSTM model took about 6–15 min of total training time, depending on the experimental setup. This trade-off is mainly due to the hybrid quantum-classical design, where quantum feature mapping and entanglement mechanisms enable the model to learn complex temporal dependencies and nonlinear relationships more effectively. Additionally, the architecture of QsLSTM helps reduce excessive computational costs, allowing for a more efficient balance between accuracy and training time compared to the deeper QLSTM configurations.

Overall, the findings confirm that the proposed QsLSTM model provides an optimal balance between computational efficiency and prediction reliability, outperforming other models. Despite slightly

Table 2

Then hyperparameter settings of different models.

Model	Ep	LB	BS	LR	D	*L	**CD	**B	***H
BiLSTM-AM [54]	200						NA		
QLSTM [55]	200	2–5	10	0.001–0.01	0.005	2	(9,2), (7,3), (5,4)	NA	NA
LEQLSTM [33]	200	10	20	0.001–0.01	0.005	2	(9, 2), (7, 3), (5, 4)	NA	NA
QsLSTM (Pro.)	200	2	65	0.001–0.01	0.005	2	(9, 2), (7, 3), (5, 4)	5	1

Notes: Ep = Epochs; LB = Lookback window; BS = Batch size; LR = Learning rate; D = Dropout rate, *L = Number of sLSTM layers, *CD = Quantum circuit design: (number of qubits, number of quantum layers), **B = Number of scalar LSTM blocks, ***H = Number of attention heads “—” indicates not available or not applicable.

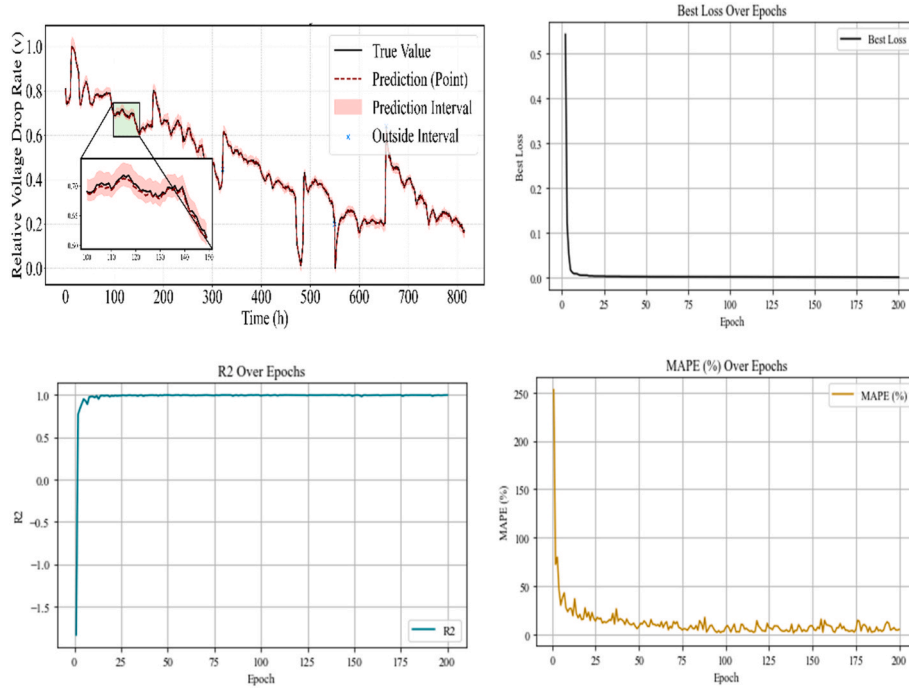


Fig. 5. Model training performances using the FC1 dataset.

higher computational demand, the gain in accuracy, coverage probability, and interpretability demonstrates its strong potential for practical time-series forecasting and prognostics applications where robust and uncertainty-aware predictions are crucial.

3.2. Relative voltage drop rate prediction using the PEM FC2 stack dataset

This section presents the relative voltage drop rate (RVDR) prediction results and associated uncertainty intervals generated by the proposed QsLSTM model, evaluated under different quantum circuit configurations using the FC2 dataset. The prediction performance is compared across two prediction horizons: $t_{sp} = 621h$ and $t_{sp} = 721h$, for BiLSTM-AM, LEQLSTM, and Transformer, as shown in Table 4 and Fig. 6(a) and (b).

The QsLSTM model consistently outperforms baseline models in RMSE across all configurations. At $t_{sp} = 621h$, the lowest RMSE (0.0016) and highest coverage (96.64 %) are achieved with the (5 qubits, 4 layers) circuit, which has a narrow NAIW of 0.1294, indicating highly accurate and confident predictions. While the LEQLSTM achieves similar RMSEs, it lacks uncertainty estimates, limiting its reliability in practical use. At $t_{sp} = 721h$, QsLSTM with a (5,4) configuration again yields the lowest RMSE (0.0014), although the coverage drops to 85.32 %. Conversely, the (7,3) and (9,3) configurations demonstrate a trade-off between slightly higher RMSEs (0.0015–0.0016) and better coverage (above 94 %), with narrower or comparable NAIW values. This suggests that deeper or broader quantum circuits can enhance interval reliability without significantly harming point prediction accuracy.

Fig. 6 (a) and (b) show the SOH prediction paths with uncertainty bands for both time horizons. The predicted intervals closely match the actual degradation trend and provide correct coverage, confirming the model's ability to generate both accurate and well-calibrated forecasts. Overall, the results show that the QsLSTM model, trained with a joint supervision loss, effectively balances forecast accuracy and uncertainty estimation. Among the tested configurations, the (5,4) setup gives the most accurate short-term forecasts, while the (7,3) and (9,3) setups

improve uncertainty calibration for longer horizons.

3.3. Remaining useful life and uncertainty limit predictions using the PEM FC2 dataset

This subsection evaluates the capability of the QLSTM model to predict the remaining useful life (RUL) of the PEMFC stack under different quantum configurations and prediction starting points (712 h and 812 h). The prediction outcomes are summarized in Table 4 and illustrated in Fig. 7(a). These results demonstrate the model's predictive accuracy and reliability under joint supervision.

At the starting point of 712 h, the proposed QsLSTM achieves highly accurate predictions across all configurations, with AE ranging from 0 to 2 h and RE under 1 %. In particular, the (7,3) configuration achieves perfect prediction (AE = 0, RE = 0 %) with 100 % RUL accuracy, outperforming the LEQLSTM baseline. Similarly, the (5,4) and (9,3) QsLSTM variants achieve AEs of 1–2 h, with RA exceeding 99.5 %.

At 812 h, the QsLSTM again shows robust performance, maintaining low AEs (0–1 h) and RE values under 1 % across all configurations. The (9,3) variant performs particularly well with only a 1-h AE and a minimal RE of 0.14 %, closely matching the actual RUL. Compared to LEQLSTM, which also performs reasonably well, the QsLSTM consistently demonstrates improved accuracy and tighter alignment with actual RUL values.

In Fig. 7(a) and (b), the QsLSTM predictions closely track the true RUL degradation path with minimal deviation. The uncertainty limits are narrow and well-calibrated, reflecting the model's ability to produce not only precise forecasts but also reliable confidence intervals.

3.4. Discussion

In summary, the QsLSTM model, trained with joint supervision loss, effectively predicts RUL with near-zero error and high reliability. The results confirm that the model is well-suited for practical PEMFC prognostics, outperforming baseline models in both short- and long-horizon

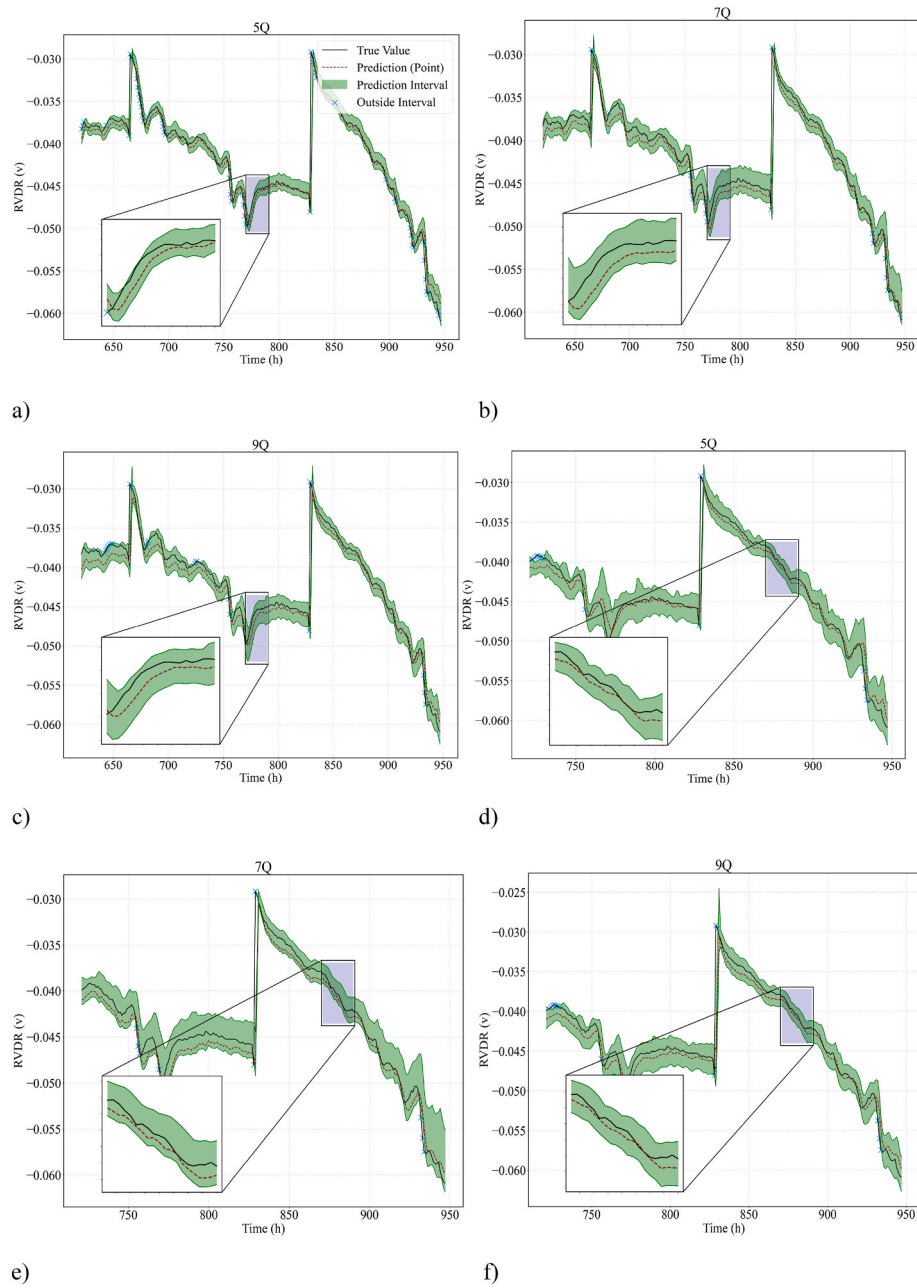


Fig. 6. The SOH estimation with uncertainty interval prediction using when $t_{sp} = 621h$ (a, b, c) and $t_{sp} = 721h$ (d, e, f).

Table 3
Comparison results of different models.

Model configurations (Qubits, Layers)		Performance metrics									
		tsp = 621 h					tsp = 721 h				
		RMSE	MAPE (%)	Cov. (%)	NAIW	Time (sec)	RMSE	MAPE (%)	Cov.(%)	NAIW	Time (sec)
(5,4)	BiLSTM-AM	0.0087	0.0635	–	–	122.74	0.0034	0.0533	–	–	243.43
	Transformer	0.0024	0.0371	–	–	160.63	0.0023	3.4546	–	–	255.89
	QLSTM	0.0067	0.1428	–	–	567.43	0.0047	0.0857	–	–	633.30
	LEQLSTM	0.0032	0.0800	–	–	311.66	0.0526	0.0029	–	–	442.92
(7,3)	QsLSTM	0.0016	0.0264	96.6360	0.1294	404.89	0.0014	0.1532	85.321	0.0560	459.55
	QLSTM	0.0075	0.1928	–	–	806.90	0.0057	0.1305	–	–	867.45
	LEQLSTM	0.0029	0.0581	–	–	358.78	0.0024	0.0520	–	–	610.22
	QsLSTM	0.0017	0.0239	95.5947	0.1110	553.19	0.0015	0.0205	94.495	0.0848	592.70
(9,3)	QLSTM	0.0081	0.2023	–	–	903.60	0.0071	0.1848	–	–	910.34
	LEQLSTM	0.0026	0.0571	–	–	769.48	0.0028	0.0453	–	–	936.94
	QsLSTM	0.0018	0.1447	90.7489	0.1173	780.54	0.0016	0.0207	94.273	0.1047	958.45

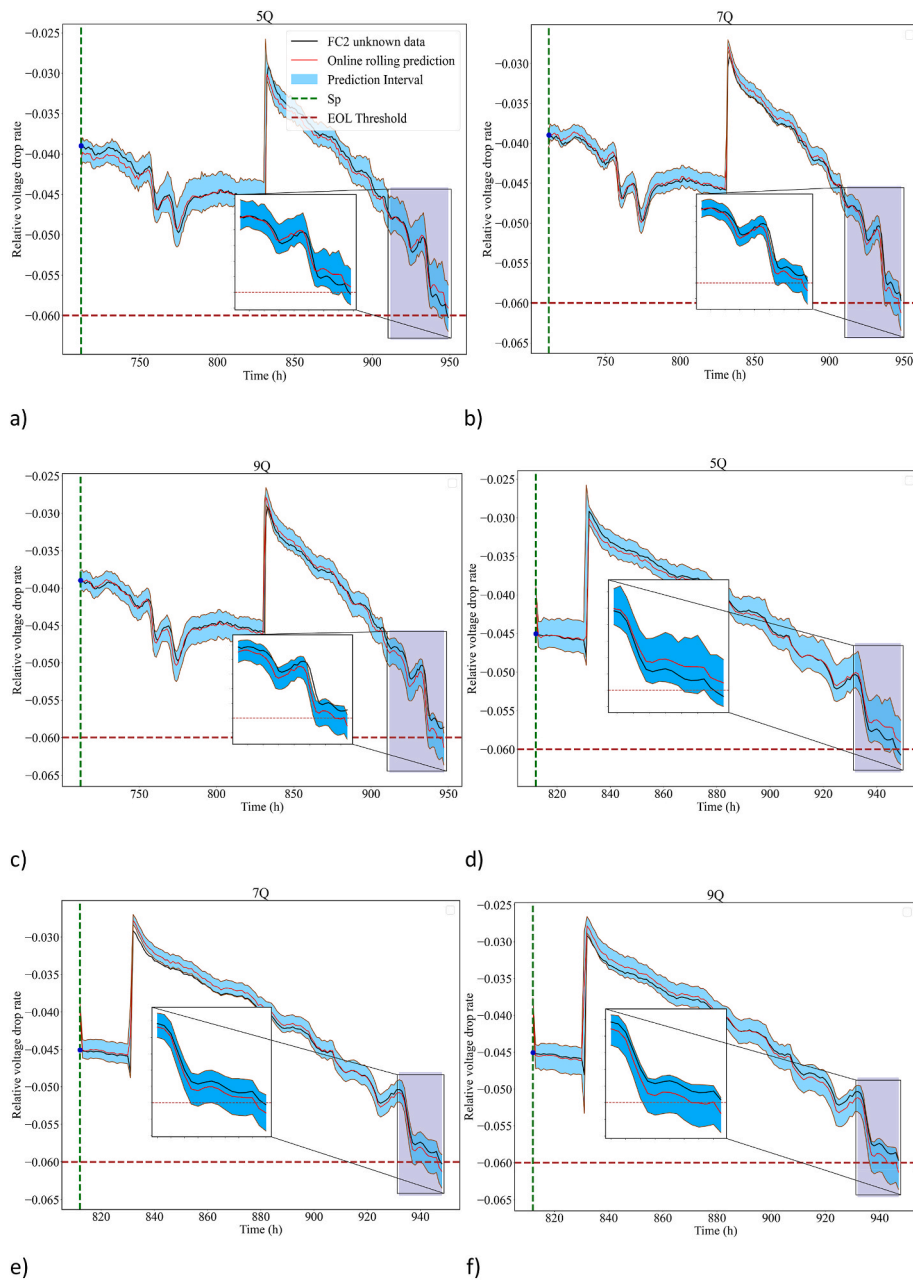
Note: QsLSTM is the proposed model.

Table 4

Results of RUL and uncertainty predictions using QsLSTM model with joint loss supervision method.

Model configurations (Qubits, Layers)		Performance metrics									
		sp = 712 h					sp = 812 h				
		Actual RUL	Pred RUL	AE	RE (%)	RA (%)	Actual RUL	Pred RUL	AE	RE (%)	RA (%)
(5,4)	LEQLSTM	224	222	2	0.89	99.11	124	122	2	1.61	98.39
	QsLSTM (Pro.)	238	239	1	0.42	99.58	138	139	1	0.72	99.28
	QLSTM	224	222	2	0.89	99.11	124	122	2	1.61	98.39
(7,3)	LEQLSTM	224	222	2	0.89	99.11	124	124	3	2.24	97.58
	QsLSTM (Pro.)	238	238	0	0	100.00	138	137	1	0.72	99.28
	QLSTM	224	222	2	0.89	99.11	124	121	3	2.42	97.58
(9,3)	LEQLSTM	224	224	0	0	100	124	124	0	0	100
	QsLSTM (Pro.)	238	236	2	0.84	99.58	138	136	1	0.14	98.55

Note: AE:, RE, RA, Pro.- Proposed.

**Fig. 7.** Future remaining useful life predictions for the FC2 stack using the QsLSTM model when Sp = 712h (a, b, c) and Sp = 812h (d, e, f).

predictions while maintaining uncertainty calibration.

4. Conclusions

This study introduces an advanced prognostic framework for proton exchange membrane fuel cells (PEMFCs) that synergistically integrates QsLSTM architecture, joint supervision for interval prediction, and transfer learning. The model effectively captures nonlinear degradation dynamics and quantifies uncertainty by learning lower, point, and upper predictive bounds. Experimental evaluations on benchmark PEMFC datasets demonstrate that QsLSTM outperforms classical BiLSTM-AM, QLSTM, LEQLSTM, and Transformer models in both SOH and RUL prediction tasks. The proposed framework achieves RMSE values as low as 0.0014 and prediction interval coverage exceeding 94 %, while maintaining narrow interval widths. Additionally, the model reliably forecasts RUL with a relative error under 1 %, making it suitable for real-time prognostics. The integration of scalar memory structures and quantum circuits enhances learning capacity without sacrificing interpretability or scalability. Using a transfer learning function, the framework generalizes effectively across different PEMFC operating profiles, addressing challenges of data scarcity and system variability. These results affirm the QsLSTM model's potential as a practical and reliable tool for predictive maintenance and performance optimization in hydrogen energy systems.

The limitations of this study include using the same type of PEMFC under two different testing conditions. More different types of PEMFC are needed for generalizing the proposed method. Additionally, increasing the number of qubits could lead to better prediction accuracy.

CRediT authorship contribution statement

Getnet Awoke Kebede: Writing – original draft, Visualization, Validation, Software, Methodology, Formal analysis, Data curation. **Fu-Kwun Wang:** Writing – review & editing, Validation, Supervision, Resources, Project administration, Methodology, Formal analysis, Conceptualization. **Bereket Haile Woldegiorgis:** Writing – review & editing, Supervision, Project administration.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgement

The second author gratefully acknowledges the financial support from the National Science and Technology Council, Taiwan, under NSTC 114-2221-E-011-116-MY3.

Data availability

The data is available on https://search-data.ubfc.fr/fclab/FR-18008901306731-2021-07-19_IIIEE-PHM-Data-Challenge-2014.html.

References

- [1] M.M.H. Bhuiyan, Z. Siddique, Hydrogen as an alternative fuel: a comprehensive review of challenges and opportunities in production, storage, and transportation, *Int. J. Hydrogen Energy* 102 (2025) 1026–1044, <https://doi.org/10.1016/j.ijhydene.2025.01.033>.
- [2] I. Sebbani, M.K. Ettouhami, M. Boulakhbar, Fuel cells: a technical, environmental, and economic outlook, *Clean Energy Syst.* 10 (2025) 100168, <https://doi.org/10.1016/j.cles.2024.100168>.
- [3] S.T.A. Rimón, M. Mourshed, M.G. Kibria, Proton exchange membrane fuel cells: advances in materials development, performance optimization, and future outlook, *Energy Convers. Manag.* X (2025) 101102, <https://doi.org/10.1016/j.ecmx.2025.101102>.
- [4] C. Acar, A. Beskese, G.T. Temur, Comparative fuel cell sustainability assessment with a novel approach, *Int. J. Hydrogen Energy* 47 (2022) 575–594, <https://doi.org/10.1016/j.ijhydene.2021.10.034>.
- [5] Z. Wang, Z. Liu, L. Fan, Q. Du, K. Jiao, Application progress of small-scale proton exchange membrane fuel cell, *Energy Rev.* 2 (2023) 100017, <https://doi.org/10.1016/j.enrev.2023.100017>.
- [6] T. Abedin, J. Pasupuleti, J.K.S. Paw, Y.C. Tak, M. Mahmud, M.P. Abdullah, M. Nur-E-Alam, Proton exchange membrane fuel cells in electric vehicles: innovations, challenges, and pathways to sustainability, *J. Power Sources* 640 (2025) 236769, <https://doi.org/10.1016/j.jpowsour.2025.236769>.
- [7] IEA, Global EV outlook 2024, IEA, Paris, *Energy Convers. Manag.* 277 (2024) 116598, 2023, <https://www.iea.org/reports/global-ev-outlook-2024>. Licence: CC BY 4.0.
- [8] S. Dirkes, J. Leidig, P. Fisch, S. Pischinger, Prescriptive lifetime management for PEM fuel cell systems in transportation applications, part I: state of the art and conceptual design, *Energy Convers. Manag.* 277 (2023) 116598, <https://doi.org/10.1016/j.enconman.2022.116598>.
- [9] Y. Wang, D.F. Ruiz Diaz, K.S. Chen, Z. Wang, X.C. Adroher, Materials, technological status, and fundamentals of PEM fuel cells – a review, *Mater. Today* 32 (2020) 178–203, <https://doi.org/10.1016/j.mattod.2019.06.005>.
- [10] Z. Liu, S. Cai, Z. Tu, S.H. Chan, Recent development in degradation mechanisms of proton exchange membrane fuel cells for vehicle applications: problems, progress, and perspectives, *Energy Storage Sav.* 3 (2024) 106–152, <https://doi.org/10.1016/j.jenss.2024.02.005>.
- [11] Z. Hua, Z. Zheng, E. Pahan, M.-C. Péra, F. Gao, A review on lifetime prediction of proton exchange membrane fuel cells system, *J. Power Sources* 529 (2022) 231256, <https://doi.org/10.1016/j.jpowsour.2022.231256>.
- [12] M. Jouin, R. Gouriveau, D. Hissel, M.-C. Péra, N. Zerhouni, Prognostics and health management of PEMFC – state of the art and remaining challenges, *Int. J. Hydrogen Energy* 38 (2013) 15307–15317, <https://doi.org/10.1016/j.ijhydene.2013.09.051>.
- [13] M.I. Gürsoy, Short and long-term prognostics of the remaining useful life of a proton exchange membrane fuel cell using deep learning and transformer model, *Int. J. Hydrogen Energy* 143 (2025) 1120–1132, <https://doi.org/10.1016/j.ijhydene.2024.12.512>.
- [14] L. Vichard, N.Y. Steiner, N. Zerhouni, D. Hissel, Hybrid fuel cell system degradation modeling methods: a comprehensive review, *J. Power Sources* 506 (2021) 230071, <https://doi.org/10.1016/j.jpowsour.2021.230071>.
- [15] J. Zuo, N.Y. Steiner, Z. Li, D. Hissel, Health management review for fuel cells: focus on action phase, *Renew. Sustain. Energy Rev.* 201 (2024) 114613, <https://doi.org/10.1016/j.rser.2024.114613>.
- [16] H.-L. Nguyen, S.-M. Lee, S. Yu, A comprehensive review of degradation prediction methods for an automotive proton exchange membrane fuel cell, *Energies* 16 (2023), <https://doi.org/10.3390/en16124772>.
- [17] Y. Dong, W. Wang, Y. Cao, M. Cao, X. Zhou, X. Hou, An improved data-driven approach and a voltage integral indicator for remaining useful life estimation of proton exchange membrane fuel cells, *J. Power Sources* 640 (2025) 236726, <https://doi.org/10.1016/j.jpowsour.2025.236726>.
- [18] C. Jia, H. He, J. Zhou, K. Li, J. Li, Z. Wei, A performance degradation prediction model for PEMFC based on bi-directional long short-term memory and multi-head self-attention mechanism, *Int. J. Hydrogen Energy* 60 (2024) 133–146, <https://doi.org/10.1016/j.ijhydene.2024.02.181>.
- [19] Z. Wang, J. Tao, Z. Liu, H. Feng, L. Ma, M. Xu, H. Su, A proton exchange membrane fuel cells degradation prediction method based on multi-scale temporal information merging network, *Energy* 313 (2024) 133995, <https://doi.org/10.1016/j.energy.2024.133995>.
- [20] C. Ferreira, G. Gonçalves, Remaining useful life prediction and challenges: a literature review on the use of machine learning methods, *J. Manuf. Syst.* 63 (2022) 550–562, <https://doi.org/10.1016/j.jmsy.2022.05.010>.
- [21] S. Wang, H. Ma, Y. Zhang, S. Li, W. He, Remaining useful life prediction method of lithium-ion batteries is based on variational modal decomposition and deep learning integrated approach, *Energy* 282 (2023) 128984, <https://doi.org/10.1016/j.energy.2023.128984>.
- [22] M. Yue, Z. Li, R. Roche, S. Jemei, N. Zerhouni, Degradation identification and prognostics of proton exchange membrane fuel cell under dynamic load, *Control Eng. Pract.* 118 (2022) 104959, <https://doi.org/10.1016/j.conengprac.2021.104959>.
- [23] T. Berghout, M. Benbouzid, Y. Amirat, Improving small-scale machine learning with recurrent expansion for fuel cells time series prognosis. University of Batna 2, Laboratory of Automation and Manufacturing Engineering, Batna, Algeria, IEEE Computer Society, 2022, <https://doi.org/10.1109/IECON49645.2022.9968566>.
- [24] Q. Liu, W. Zang, W. Zhang, Y. Zhang, Y. Tong, Y. Feng, Steady-state model enabled dynamic PEMFC performance degradation prediction via recurrent neural network, *Energies* 18 (2025), <https://doi.org/10.3390/en18102665>.
- [25] S. Zhang, T. Chen, F. Xiao, R. Zhang, Degradation prediction model of PEMFC based on multi-reservoir echo state network with mini reservoir, *Int. J. Hydrogen Energy* 47 (2022) 40026–40040, <https://doi.org/10.1016/j.ijhydene.2022.09.160>.
- [26] Y.B. Yakut, Short-term voltage loss prediction in PEMFCs using single-step and multi-step LSTM models, *Int. J. Hydrogen Energy* (2025), <https://doi.org/10.1016/j.ijhydene.2025.02.317>.
- [27] Z. Zhang, Y.-X. Wang, H. He, F. Sun, A short- and long-term prognostic associating with remaining useful life estimation for proton exchange membrane fuel cell, *Appl. Energy* 304 (2021), <https://doi.org/10.1016/j.apenergy.2021.117841>.

- [28] B. Long, K. Wu, P. Li, M. Li, A novel remaining useful life prediction method for hydrogen fuel cells based on the gated recurrent unit neural network, *Appl. Sci.* 12 (2022), <https://doi.org/10.3390/app12010432>.
- [29] W. Huo, W. Li, Z. Zhang, C. Sun, F. Zhou, G. Gong, Performance prediction of proton-exchange membrane fuel cell based on convolutional neural network and random forest feature selection, *Energy Convers. Manag.* 243 (2021) 114367, <https://doi.org/10.1016/j.enconman.2021.114367>.
- [30] Z. Tao, C. Zhang, J. Xiong, H. Hu, J. Ji, T. Peng, M.S. Nazir, Evolutionary gate recurrent unit coupling convolutional neural network and improved manta ray foraging optimization algorithm for performance degradation prediction of PEMFC, *Appl. Energy* 336 (2023), <https://doi.org/10.1016/j.apenergy.2023.120821>.
- [31] C. Zhang, T. Niu, Z. Wei, X. Hu, F. Pan, T. Zeng, Y. Si, C.S. Chin, Lifespan prediction model for proton exchange membrane fuel cell vehicle based on time series information feature extraction and optimization, *Renew. Energy* 243 (2025) 122558, <https://doi.org/10.1016/j.renene.2025.122558>.
- [32] S. Li, W. Luan, C. Wang, Y. Chen, Z. Zhuang, Degradation prediction of proton exchange membrane fuel cell based on Bi-LSTM-GRU and ESN fusion prognostic framework, *Int. J. Hydrogen Energy* 47 (2022) 33466–33478, <https://doi.org/10.1016/j.ijhydene.2022.07.230>.
- [33] G.A. Kebede, S.-C. Lo, F.-K. Wang, J.-H. Chou, Transfer learning-based deep learning models for proton exchange membrane fuel remaining useful life prediction, *Fuel* 367 (2024) 131461, <https://doi.org/10.1016/j.fuel.2024.131461>.
- [34] F.-K. Wang, G.A. Kebede, S.-C. Lo, B.H. Woldegiorgis, An embedding layer-based quantum long short-term memory model with transfer learning for proton exchange membrane fuel stack remaining useful life prediction, *Energy* 308 (2024) 133054, <https://doi.org/10.1016/j.energy.2024.133054>.
- [35] J.-H. Chou, F.-K. Wang, Future prediction on the remaining useful life of proton exchange membrane fuel using temporal fusion transformer model, *Renew. Energy* 247 (2025) 123019, <https://doi.org/10.1016/j.renene.2025.123019>.
- [36] L. Tian, Y. Gao, H. Yang, R. Wang, Multi-scenario long-term degradation prediction of PEMFC based on generative inference informer model, *Appl. Energy* 377 (2025) 124398, <https://doi.org/10.1016/j.apenergy.2024.124398>.
- [37] Y. Xu, J. Wang, D. Hu, D. Lu, X. Zhang, W. Wei, H. Ding, H. Zhang, BL-DA transformer lifespan degradation prediction model of fuel cell using relative voltage loss rate health indicator, *World Electr. Veh. J.* 16 (2025), <https://doi.org/10.3390/wevj16060290>.
- [38] Z. Hua, Z. Zheng, E. Pahan, M.-C. Péra, GaoF. Remaining useful life prediction of PEMFC systems under dynamic operating conditions, *Energy Convers. Manag.* 231 (2021) 113825, <https://doi.org/10.1016/j.enconman.2021.113825>.
- [39] N.D. Nguyen, V.T. Nguyen, Q.T.H. Nghiem, J. Woo, Y. Kim, S. Yu, In-situ measurement of RH/T distribution in the straight-channel PEMFC under long-term operation with NEDC mode, *Appl. Energy* 392 (2025) 125994, <https://doi.org/10.1016/j.apenergy.2025.125994>.
- [40] X. Meng, M. Liu, J. Mei, X. Li, S. Grigoriev, H.M. Hasanien, X. Tang, R. Li, C. Sun, Polarization loss decomposition-based online health state estimation for proton exchange membrane fuel cells, *Int. J. Hydrogen Energy* 157 (2025) 150162, <https://doi.org/10.1016/j.ijhydene.2025.150162>.
- [41] Y. Yang, X. Yu, W. Zhu, C. Xie, B. Zhao, L. Zhang, Y. Shi, L. Huang, R. Zhang, Degradation prediction of proton exchange membrane fuel cells with model uncertainty quantification, *Renew. Energy* 219 (2023), <https://doi.org/10.1016/j.renene.2023.119525>.
- [42] X. Yu, Y. Yang, C. Xie, Y. Li, B. Zhao, L. Zhang, X. Yu, et al., Remaining-useful-lifetime prediction of proton exchange membrane fuel cell considering model uncertainty quantification on the full-time scale, *IEEE Trans. Electr. Inf. Technol.* 10 (3) (2024) 7443–7455, <https://doi.org/10.1109/TTE.2023.3336324>.
- [43] W. Zhu, B. Guo, Y. Li, Y. Yang, C. Xie, J. Jin, H.B. Gooi, Uncertainty quantification of proton-exchange membrane fuel cells degradation prediction based on bayesian-gated recurrent unit, *eTransportation* 16 (2023) 100230, <https://doi.org/10.1016/j.etran.2023.100230>.
- [44] Y. Yu, C. Hu, X. Si, J. Zheng, J. Zhang, Averaged Bi-LSTM networks for RUL prognostics with non-life-cycle labeled dataset, *Neurocomputing* 402 (2020) 134–147, <https://doi.org/10.1016/j.neucom.2020.03.041>.
- [45] W. Zhu, B. Guo, Y. Li, Y. Yang, C. Xie, J. Jin, Y. Si, C.S. Chin, Lifespan prediction model for proton exchange membrane fuel cell vehicle based on time series information feature extraction and optimization, *Renew. Energy* 16 (2025) 122558, <https://doi.org/10.1016/j.renene.2025.122558>.
- [46] X. Meng, C. Sun, J. Mei, X. Tang, H.M. Hasanien, J. Jiang, F. Fan, K. Song, Fuel cell life prediction considering the recovery phenomenon of reversible voltage loss, *J. Power Sources* 625 (2025) 235664, <https://doi.org/10.1016/j.jpowsour.2024.235664>.
- [47] Q.A. Memon, M. AlAhmad, M. Pecht, Quantum computing: navigating the future of computation, challenges, and technological breakthroughs, *Quantum Rep.* 6 (2024) 627–663, <https://doi.org/10.3390/quantum6040039>.
- [48] S.Y.C. Chen, S. Yoo, Y.L.L. Fang, Quantum long short-term memory. ICASSP, in: *IEEE Int Conf Acoust Speech Signal Process - Proc 2022*, 2022-May, pp. 8622–8626, <https://doi.org/10.1109/ICASSP43922.2022.9747369>.
- [49] S.Z. Khan, N. Muzammil, S. Ghafoor, H. Khan, S.M.H. Zaidi, A.J. Aljohani, I. Aziz, Quantum long short-term memory (QLSTM) vs. classical LSTM in time series forecasting: a comparative study in solar power forecasting, *Front. Phys.* 12 (2024).
- [50] Hissel D, Turpin C, Roziere J, Harel F. IEEE PHM 2014 data challenge. <https://doi.org/10.25666/DATAUBFC-2021-07-19>.
- [51] Y. Li, D.-I. Stroe, Y. Cheng, H. Sheng, X. Sui, R. Teodorescu, On the feature selection for battery state of health estimation based on charging–discharging profiles, *J. Energy Storage* 33 (2021) 102122, <https://doi.org/10.1016/j.est.2020.102122>.
- [52] K. Song, T. Hou, J. Jiang, S.A. Grigoriev, F. Fan, J. Qin, Z. Wang, C. Sun, Thermal management of liquid-cooled proton exchange membrane fuel cell: a review, *J. Power Sources* 648 (2025) 237227, <https://doi.org/10.1016/j.jpowsour.2025.237227>.
- [53] M. Beck, K. Pöppel, M. Spanring, A. Auer, O. Prudnikova, M. Kopp, et al., xLSTM: extended long short-term memory, 1–55, <https://doi.org/10.48550/arXiv.2405.04517>, 2024.
- [54] Kiani ShahvandiM, SojaB. GouJ, Deep quantum learning with long short-term memory for geodetic time series prediction: application to length of day prediction, *Earth Space Sci. Open Arch.* 1 (2021).
- [55] D. Li, Multivariate time series prediction based on quantum enhanced LSTM models, 1271920:20, <https://doi.org/10.1117/12.2685468>, 2023.
- [56] C. Dong, D. Sun, Multi-source domain transfer learning with small sample learning for thermal runaway diagnosis of lithium-ion battery, *Appl. Energy* 365 (2024) 123248, <https://doi.org/10.1016/j.apenergy.2024.123248>.